

1 Ideal Over-the-Air Computation

Before introducing the practical impairments inherent to wireless channels, such as Inter-Symbol Interference (ISI), multipath fading and symbol to filter mismatch, it is essential to establish a baseline model of Over-the-Air (OTA) computation under ideal conditions. In this scenario, the wireless medium is assumed to be perfectly synchronized, lacking frequency-selective fading, and corrupted only by Additive White Gaussian Noise (AWGN). This chapter aggregates the mathematical formulation, MATLAB code, Simulink models, and GNU Radio flowgraphs that demonstrate this ideal baseline.

1.a Mathematical Formulation of the Ideal Channel

Consider a wireless sensor network consisting of K End Device (ED) transmitting analog values to a Fusion Center (FC). Let $x_k(t)$ denote the continuous-time signal transmitted by the k -th device. In an ideal Multiple Access Channel (MAC) where perfect Channel State Information (CSI) allows for perfect pre-equalization of phase and amplitude shifts, the signals superimpose perfectly at the receiver.

The received baseband signal $y(t)$ at the FC is mathematically expressed as:

$$y(t) = \sum_{k=1}^K x_k(t) + n(t) \quad (1)$$

where $n(t) \sim \mathcal{N}(0, \sigma^2)$ represents the AWGN. Because there is no multipath delay spread, the channel impulse response is effectively a Dirac delta function $h(t) = \delta(t)$, meaning the current symbol is entirely unaffected by past or future symbols.

1.b MATLAB Simulation and Code Analysis

To validate this superposition principle, the ideal environment can be modeled using MATLAB. The following script generates normally distributed simulated sensor readings, aligns them perfectly in time, and adds AWGN to emulate the ideal MAC.

Listing 1: MATLAB script for ideal OTA computation aggregation.

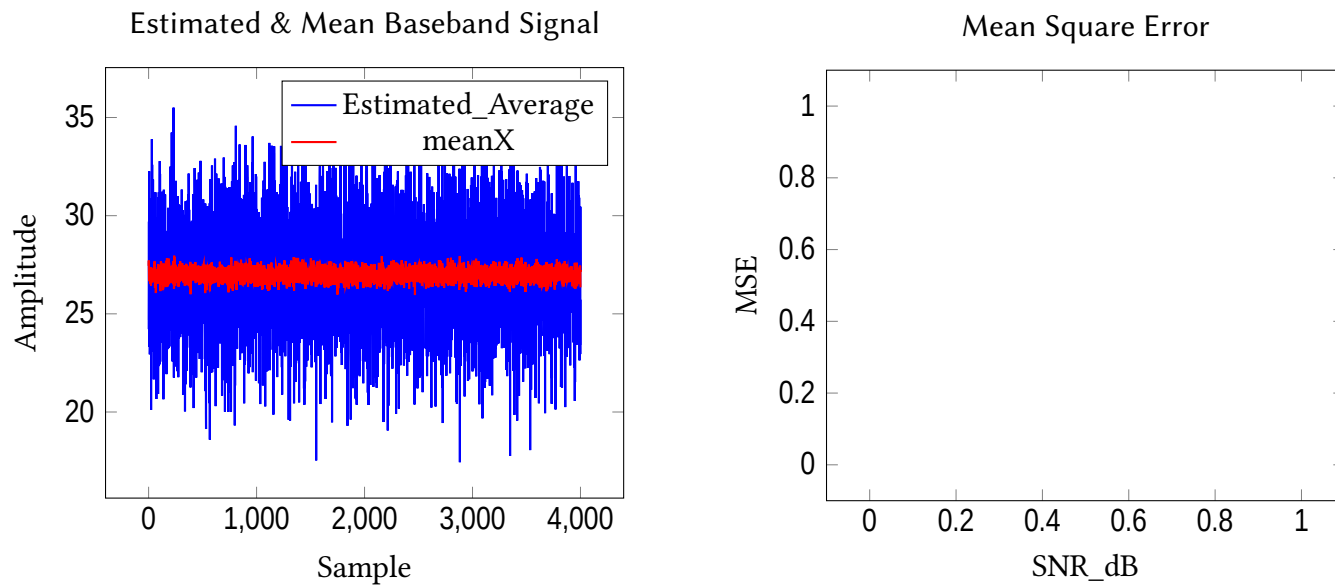
```

1      % Ideal OTA Computation Simulation
2      K = 3;                      % Number of sensors
3      N = 1000;                  % Number of samples
4      SNR_dB = 20;               % Signal-to-Noise Ratio
5
6      % Generate random temperature readings around 27 degrees Celsius
7      X = 27 + randn(K, N);
8
9      % The Air acts as an ideal adder (MAC Superposition)
10     Ideal_Sum = sum(X, 1);
11
12     % Add AWGN

```

```
18 % Display Mean Squared Error (MSE)
19 MSE = immse(Estimated_Average, mean(X, 1)); fprintf('Ideal Baseline MSE
    at %d dB SNR: %f\n', SNR_dB, MSE);
```

As demonstrated by the script, when synchronization is perfect, the MSE is strictly a function of the AWGN variance.



On the 1.b plot how much the AWGN influences the variance of the estimated received signal and the average values of the randomly generated signal.

```
[ title=3D MSE Surface Plot (sps = 2), xlabel=SNR (dB), ylabel=Roll-off ( ), zlabel=MSE, view=6030,
  grid=on, colorbar, colormap=jet, ] 3[surf, mesh/cols=21] table[ x=SNR_dB , y=Roll-off , z=MSE ]
```

Figure 1: Mean Squared Error variation across SNR and Roll-off factors.

1.c MATLAB Simulink Architecture

Transitioning from raw code to a block-level architecture, MATLAB Simulink provides a clear visual representation of the feedforward ideal system. In this model, the transmitters use standard Pulse Amplitude Modulation (PAM) without the need for complex feedback loops or equalizers.

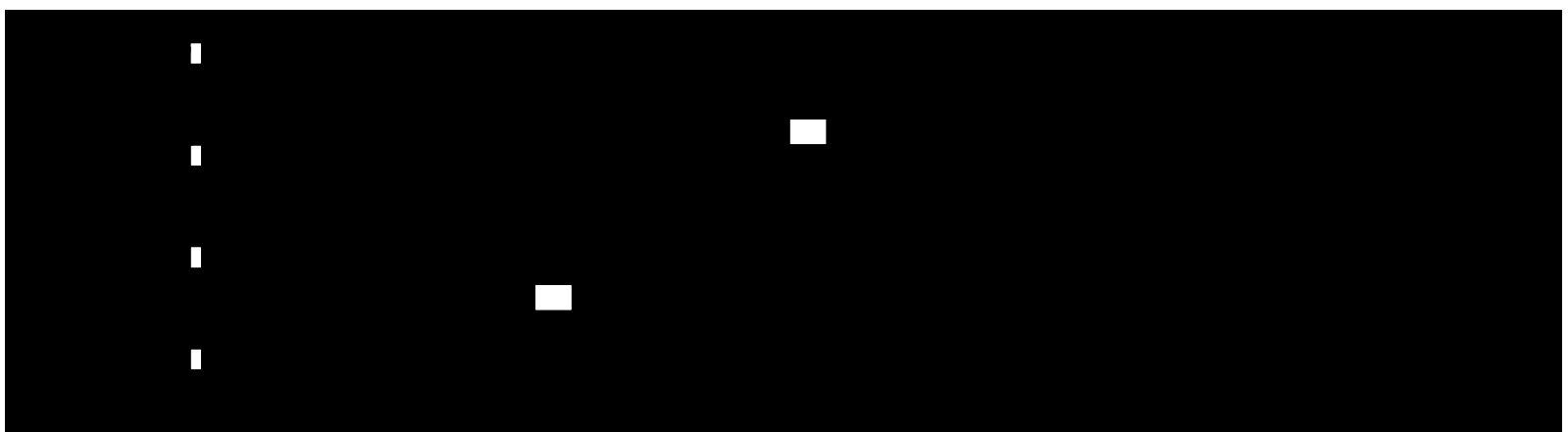


Figure 2: Simulink block diagram representing the ideal OTA computation network over an AWGN channel.

Observing the Mean Squared Error (MSE) calculation scope yields how apart the the estimate can be because of the AWGN.

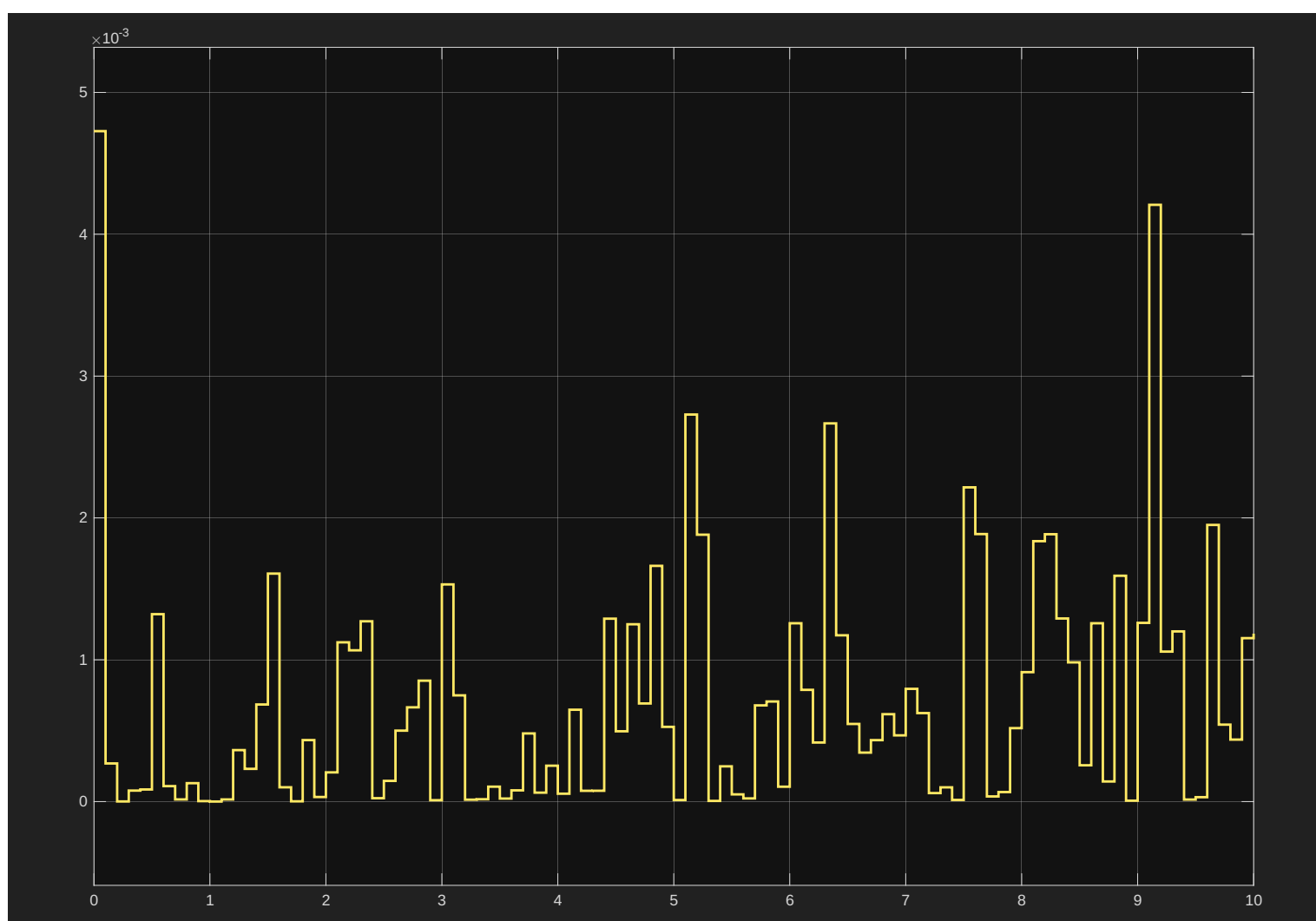


Figure 3: Mean Square Error of the ideal OTA computation network over an AWGN channel.

1.d GNU Radio Validation

Finally, the ideal model is verified using a Software Defined Radio (SDR) environment in GNU Radio. To maintain the ideal conditions mathematically defined in (1), the Root Raised Cosine (RRC) filters are perfectly matched. Specifically, the interpolation rate at the transmitters and the decimation rate at the receiver are identically matched to the symbol design rate ($\text{sps_actual} = \text{sps_design} = 6$). In other words the upsampling rate of the Finite Impulse Response (FIR) filters matches the spacing between sample generation.

What is interesting to note about the flowgraph implementation is that because no digital modulation technique is implemented, downsampling is mandatory. Each device for every one symbol it transmits sixteen are discarded in order for pulse shaping to work.

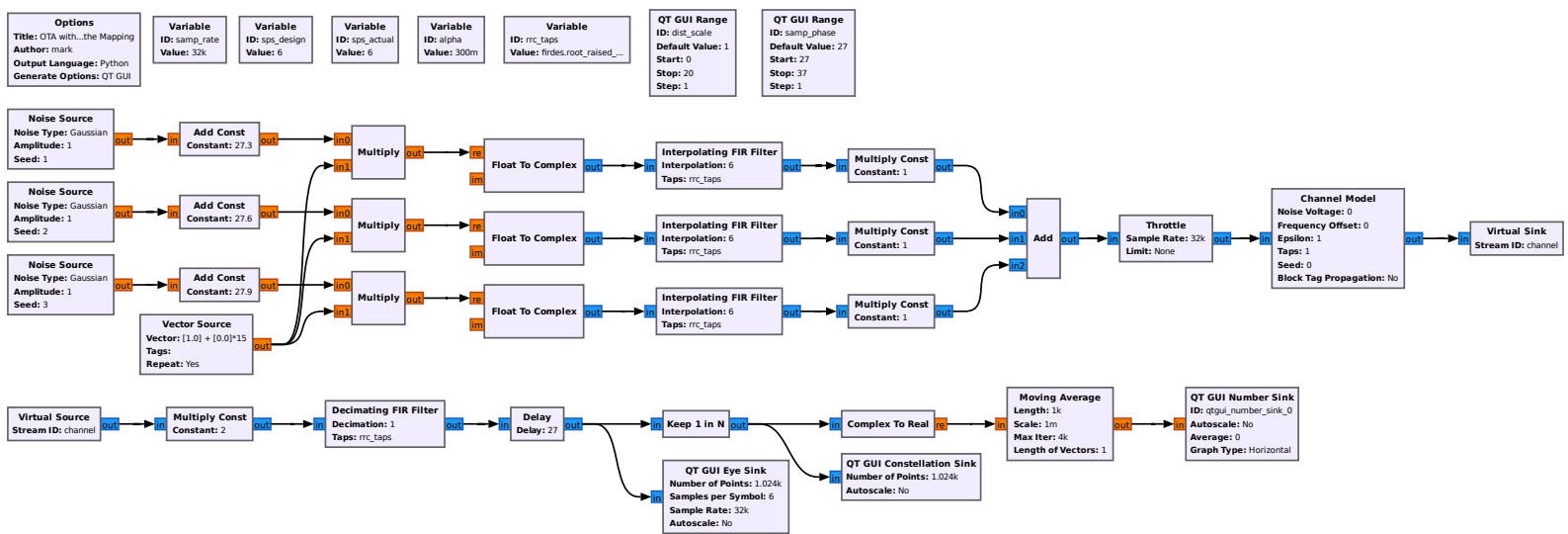


Figure 4: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

Under these constraints, the Eye Diagram remains perfectly "open" at the optimal sampling phase, and the Constellation Diagram exhibits tightly clustered points. The absence of horizontal dispersion in the constellation confirms that, prior to the introduction of timing offsets or Faster-Than-Nyquist signaling, the MAC superposition principle flawlessly computes the arithmetic average of the agricultural sensors.

What is interesting to observe is that the sampling phase needs to be set at 33 in order to receive the maximum available data. That is because at sample 33 out of 96¹ is the physical "group delay" caused by the math inside Interpolating and Decimating FIR filters.

¹The cycle is 16 symbols long. Since there are 6 samples per symbol, your total cycle is 96 samples (16 * 6).

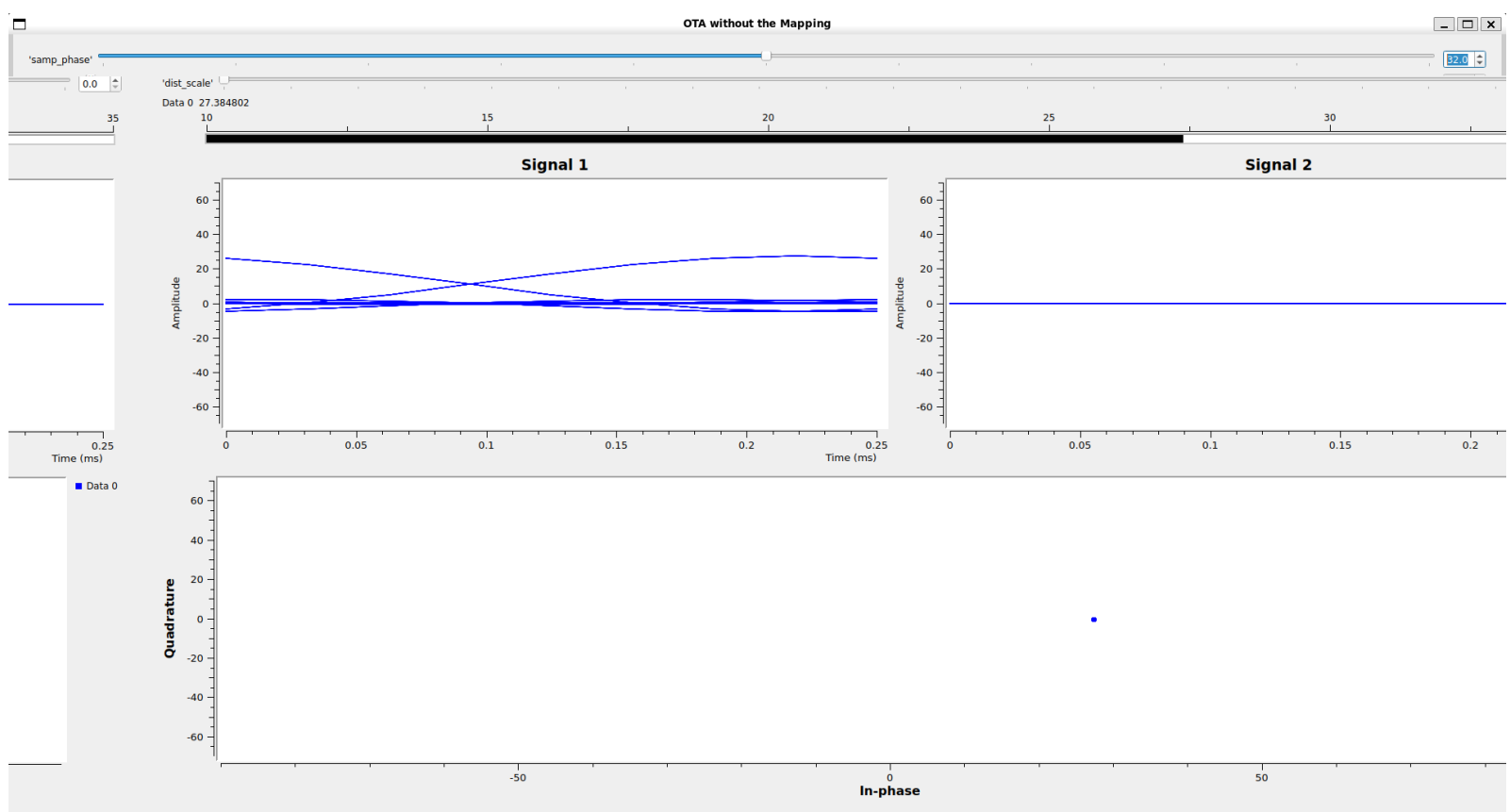


Figure 5: GNU Radio Eye & Constellation Diagram for the ideal, Nyquist-compliant scenario.